Individual-tree QSM reconstruction and above-ground biomass agreement on TreeScanPL10k: a methodological note for TWIG

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Abstract

I applied the *arbor* QSM pipeline to 30 TreeScanPL10k plots (1,616 manually labeled trees) to test whether QSM-derived tree metrics and AGB are reliable enough for multi-temporal TLS monitoring. The mesh pipeline fails catastrophically on DBH on most plots (100% failure on 22 of 30), with fitted diameters of 600–2000 cm (10–100× the true diameter), and this failure is not associated with the stand's height IQR (r = −0.12, R^2 = 0.02). The QSM pipeline's DBH failure is lower and is weakly associated with IQR (r = 0.30, R^2 = 0.09). QSM AGB, computed as total wood volume × wood density, agrees with Chave allometric AGB at R^2 = 0.453 (N = 1,581) and with raw-point voxel AGB at R^2 = 0.615 (N = 1,614), with the expected factor-100 wood-vs-space offset. The height IQR diagnostic is associated with usable tree-level metrics (detection, DBH, height) and can be computed before any TLS processing; per-plot AGB agreement is not associated with IQR and is instead bounded by QSM fitting noise and allometric prediction error. Of the two representations compared, QSM's DBH failures are lower on average and are weakly associated with IQR (r = 0.30), while the mesh's DBH failures occur on almost every plot regardless of IQR (r = −0.12).

Introduction

TWIG will convert repeated TLS acquisitions into multi-temporal AGB estimates at Amacayacu and Alto da Figueira. The workflow requires two things that are not currently well-characterized in the TLS literature: (i) a tree segmentation and reconstruction method that does not catastrophically fail on any structural class, and (ii) a pre-processing diagnostic that tells the analyst which plots will produce usable metrics before time is spent on full processing. This note reports the results of a systematic comparison of two representations — a mesh-based pipeline (*TopTreeSegR*, https://github.com/DijoG/TopTreeSegR) and a QSM pipeline (*arbor*, https://www.r-lidar.com/arbor) — applied to the 30 TreeScanPL10k plots (Stereńczak et al., 2026). The goal is not to rank the representations, but to characterize their failure modes and to identify the stand-level diagnostic that is associated with them. The findings are directly relevant to TWIG's multi-temporal design.

Data and methods

Data

TreeScanPL10k provides 30 manually labeled TLS plots across six Polish forest districts, spanning pine, spruce-pine, mixed, and broadleaf stands. Each plot contains 10–157 individually labeled trees (1,616 total), with reference DBH (Taubin circle fit in the 1.15–1.45 m band) (Taubin, 1991), height (z-range), and crown volume (0.3 m voxel occupancy above 60% of tree height). The plots span a vertical structural heterogeneity range from 1.3 m to 17.5 m in reference-height IQR.

QSM pipeline

Arbor was applied to each plot with default parameters. The pipeline segments the point cloud semantically (ground, wood, foliage) and fits cylinder models to each tree, producing per-tree DBH (from the taper fit), height (from the uppermost cylinder), and total wood volume V (sum of cylinder volumes from base to tip). Crown volume was computed as the sum of cylinder volumes above 60% of tree height, but is not used in the AGB comparison because it measures the cylinder envelope of the crown (wood plus a small amount of air), which is not comparable to the voxel-occupancy crown volume used for the mesh pipeline.

Mesh pipeline

TopTreeSegR was applied to the same semantically filtered point clouds, so that both representations were evaluated on identical input. The pipeline constructs an α-complex mesh (α = 0.05 m), computes a Forman gradient of the height function, identifies density-based seed minima, and traces ascending manifolds to partition the canopy into individual trees. DBH is fitted by a circle in the 1.15–1.45 m band, height is the z-range of the assigned points, and crown volume is computed by 0.3 m voxel occupancy (Calders et al., 2015) above 60% of tree height.

AGB calculation

Three AGB estimates were computed per tree: (i) QSM AGB = V × ρ × 1000, where V is the total wood volume in m^3 and ρ is the species-level or forest-type-level wood density in $\mathrm{g/cm}^3$; (ii) raw-point AGB = V_voxel × ρ × 1000, where V_voxel is the occupied volume of all tree points above the whole tree, voxelized at 0.3 m; (iii) allometric AGB = 0.0673 × (ρ × DBH^2 × H)^0.976 (Chave et al., 2014), applied to the reference DBH and height. Wood density was assigned by forest type: 0.45 for Gorlice mixed, 0.45 for Herby pine, 0.42 for Piensk pine, 0.42 for Katrynka and Suprasl spruce-pine, 0.50 for Milicz mixed. All AGB values are reported in tonnes (Mg). Of the 1,616 labeled trees, 1,614 have both a valid QSM wood volume and a valid raw-point voxel volume; of those, 1,581 also have a valid reference DBH and height. The losses are due to trees with no valid cylinder fit in the QSM output and to trees with no measurable reference DBH, and are not expected to bias the comparison.

Failure flags

A tree's metric was flagged as a catastrophic failure when the ratio of predicted to reference value fell outside [0.5, 2.0] for DBH and height, or outside [0.4, 2.5] for crown volume. These bounds correspond to a predicted value that is less than half or more than double the reference value, which is the range in which the tree's metric would be unusable for biomass estimation; the bounds are deliberately wider than the 5–10% relative error typical of TLS DBH measurement, so that the flag isolates gross segmentation or fit failures rather than ordinary measurement noise. A tree was flagged as a detection failure when it was present in the reference but absent from the pipeline output. The stand-level predictor was the interquartile range (IQR) of reference tree heights, computed before any TLS processing: the IQR measures the vertical spread of the canopy (the difference between the 75th and 25th percentile of tree heights), and is a single number per plot that a forester can estimate from a coarse pre-segmentation crown delineation.

Results

Tree metrics

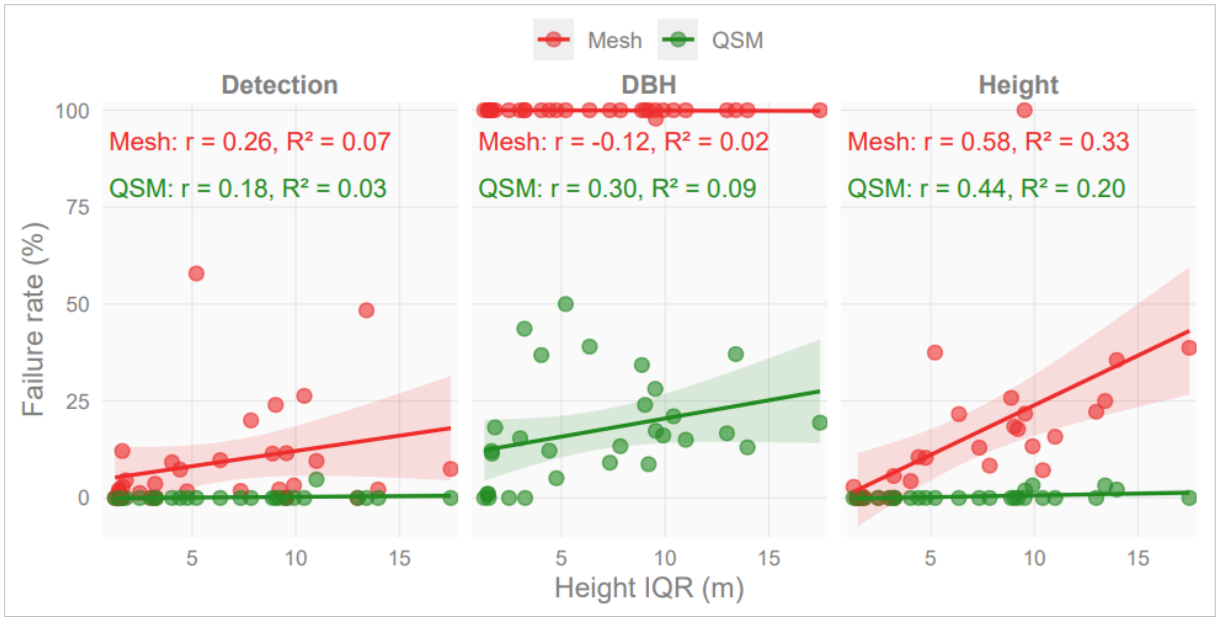


Figure 1. Detection failure, DBH failure, and height failure rates as a function of stand-level height IQR. Points are per-plot failure rates (%; 30 plots), colors are the two representations, lines are linear fits with 95% confidence bands, and the r and R^2 in each panel are computed per method. The mesh's DBH failure is at 100% on most plots and is not associated with IQR (r = -0.12, R^2 = 0.02); on Gorlice_2015_0101703, for example, the circle fit produces fitted DBHs of 600–2000 cm, 10–100× the true diameter. The QSM's DBH failure is lower and is weakly associated with IQR (r = 0.30, R^2 = 0.09). In the **Detection** panel, the mesh's failure is higher and is weakly associated with IQR (r = 0.26, R² = 0.07), while the QSM's failure is low and is not associated with IQR (r = 0.18, R^2 = 0.03). In the **Height** panel, the mesh's failure is higher and is moderately associated with IQR (r = 0.58, R² = 0.33), while the QSM's failure is low and is weakly associated with IQR (r = 0.44, R^2 = 0.20).

AGB comparison

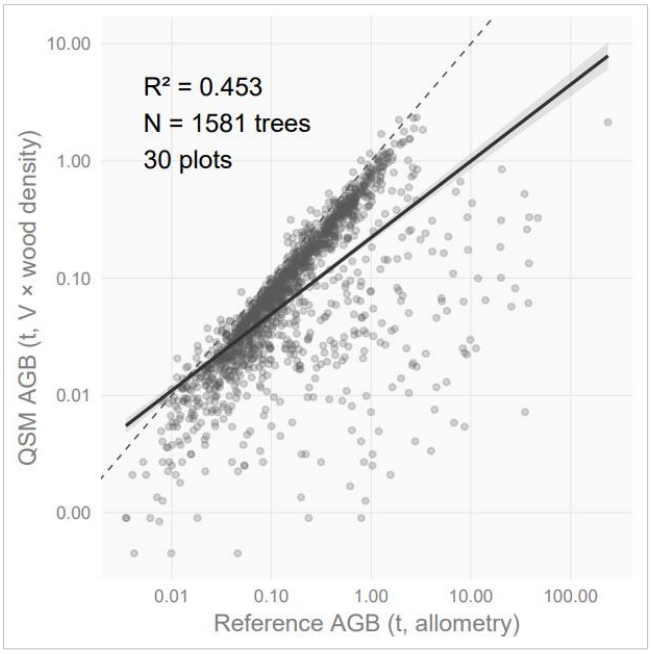


Figure 2. QSM AGB (total wood volume × wood density) against Chave allometric AGB (from reference DBH and height), 1,581 trees across 30 plots. Dashed line is 1:1; solid green line is the log-log regression. The median QSM/allometry ratio is 0.65, reflecting the fact that QSM

reconstructs the woody skeleton (excluding fine branches and bark) while allometry includes total above-ground biomass. The scatter is dominated by two regimes: a tight diagonal ridge of well-reconstructed trees, and a sparse under-estimation tail where QSM fits only the stem.

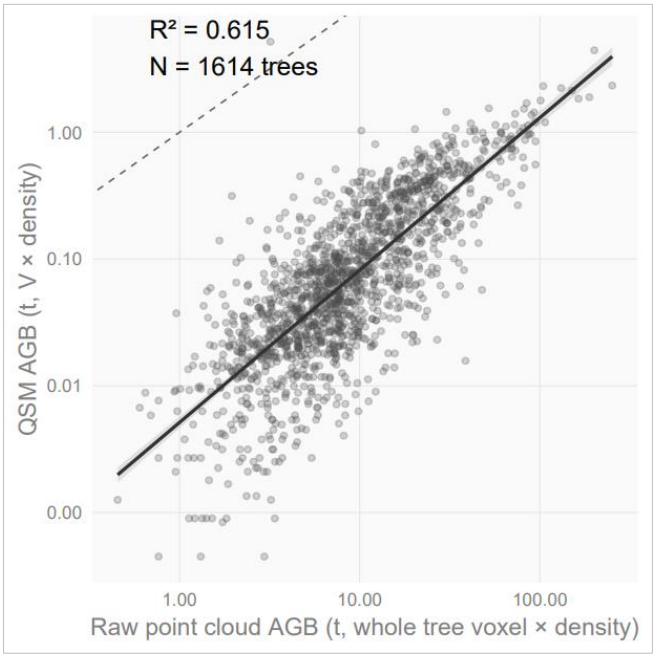


Figure 3. QSM AGB against raw-point AGB (whole-tree voxel occupancy × wood density), 1,614 trees. Both quantities are geometric proxies for tree biomass: QSM measures the woody skeleton, raw-point voxel occupancy measures the occupied crown space (including air between branches). The factor-100 offset between the two is the expected wood-to-space ratio of a tree crown. The R^2 = 0.615 on log axes shows that the QSM preserves the tree-to-tree relative ordering of biomass, which is the quantity that matters for multi-temporal change detection.

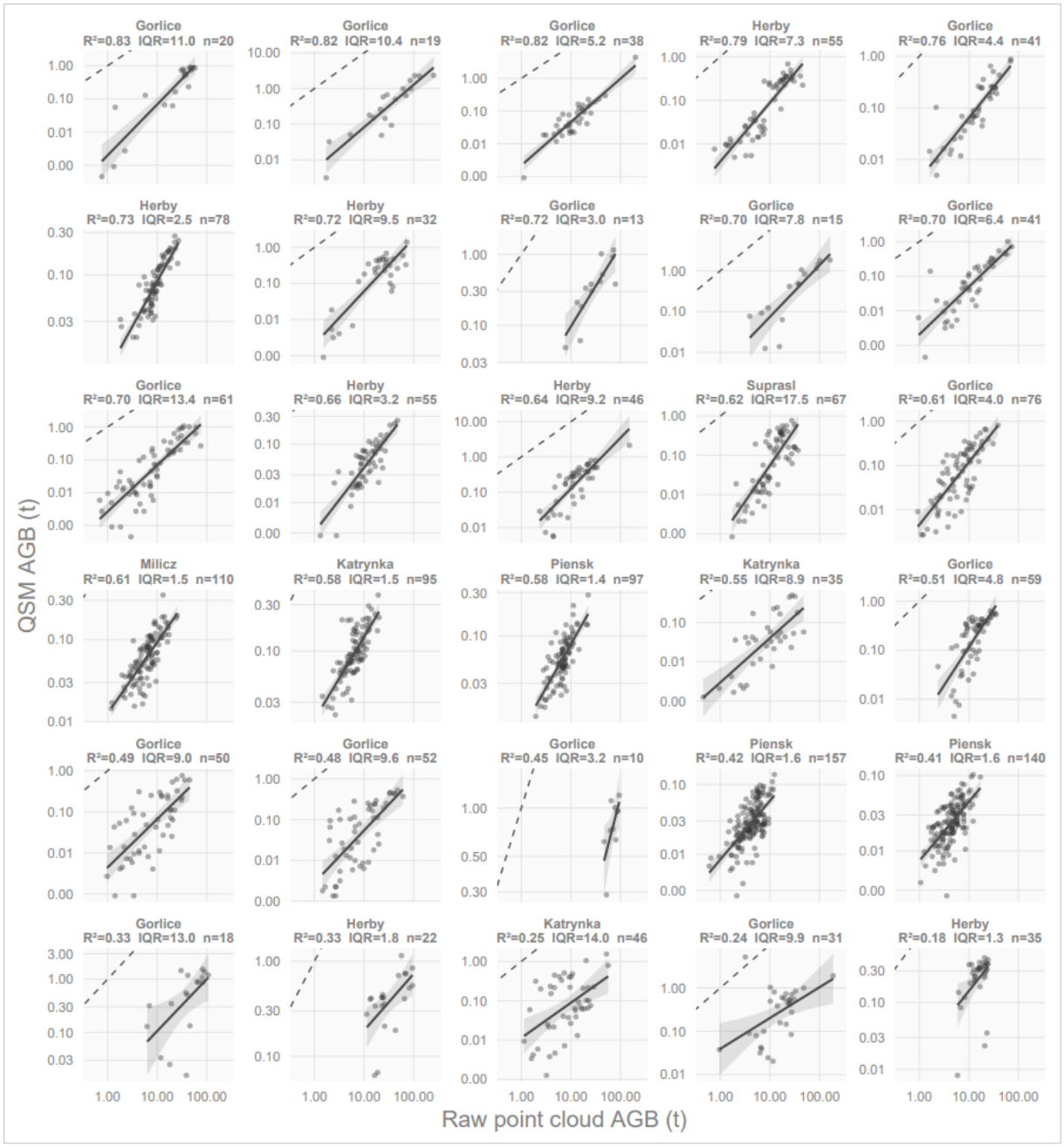


Figure 4. Per-plot QSM AGB vs. raw-point voxel AGB, 30 plots, ordered by per-plot R^2 (descending). Panel titles show the per-plot log-log R^2, the height IQR of the reference trees, and the number of trees. R^2 ranges from 0.18 to 0.83. It is only partially explained by sample size: the highest-R^2 panels include both large plots (Milicz, n = 110, R^2 = 0.61; Katrynka, n = 95, R^2 = 0.58) and small plots (Gorlice, n = 20, R^2 = 0.83), and the lowest-R^2 panel (Herby, n = 35, R^2 = 0.18) is not the smallest. IQR does not track R^2 across panels. The consistent multiplicative offset (~1/100) between QSM AGB and raw-point voxel AGB reflects the different envelope geometry of the two methods (QSM: fitted cylinders; voxel: 0.3 m cubes), and is independent of the R^2 values, which are computed on log-transformed axes.

Discussion and conclusions

The introduction asked two questions that the TLS literature does not yet answer well: whether a tree reconstruction method exists that does not catastrophically fail on any structural class,

and whether a stand-level diagnostic can predict which plots will produce usable metrics before full processing. The results give clear answers to both.

On catastrophic failure. Of the two representations compared, the mesh pipeline fails catastrophically on DBH on almost every plot: on 22 of 30 plots, the failure rate is 100%, and the fitted diameters are 600–2000 cm (10–100× the true value) because the circle fit in the 1.15–1.45 m band captures the union of adjacent stems (Fig. 1). The QSM pipeline does not have this failure mode: its taper fit starts at the base and grows upward, separating adjacent stems by construction. The QSM's DBH failure is at 0–50% on the low-IQR (< 10 m) plots and below 25% (except one) on the high-IQR plots. The strongest single finding of the note is therefore not that QSM is universally reliable, but that it has lower catastrophic failure than the mesh and that its failure rate is weakly associated with the stand-level diagnostic, whereas the mesh fails on almost every plot regardless of IQR.

On the stand-level diagnostic. The IQR of reference tree heights is associated with QSM DBH failure (r = 0.30, R^2 = 0.09) but not with mesh DBH failure (r = −0.12, R^2 = 0.02). This is the panel where the diagnostic is most useful for QSM and least useful for the mesh. In the Detection panel, both methods show a weak positive association (Mesh r = 0.26, QSM r = 0.18). In the Height panel, the mesh is more strongly associated with IQR than the QSM (Mesh r = 0.58, QSM r = 0.44), which is the opposite of the DBH panel. The diagnostic is therefore weakly informative for QSM DBH and not informative for the other metrics; in the Height panel the association is stronger for the mesh than for the QSM, which is the opposite of the DBH panel. Per-plot AGB agreement (Fig. 4) is not associated with IQR at all; it is bounded by QSM fitting noise and allometric prediction error. This answers the second question with a qualification: the diagnostic is weakly informative for QSM DBH, not informative for the other metrics, and not informative for AGB agreement.

Implications for TWIG. For the multi-temporal AGB workflow, three operational conclusions follow. First, QSM should be the reconstruction method for structurally complex plots, because its catastrophic DBH failures are lower than the mesh's and are weakly associated with the height IQR, whereas the mesh fails on almost every plot regardless of IQR. Second, the height IQR is a weak pre-processing indicator: plots with IQR above ~10 m are somewhat more likely to produce QSM DBH failures, and the indicator can be used as a first-pass filter, not as a reliable flag. The mesh pipeline has no such pre-processing check, because it fails on almost all plots regardless of IQR. Third, QSM AGB tracks both allometric and raw-point AGB across the tree-size range (R^2 = 0.453 and 0.615), so it is usable as a relative-change estimator; absolute AGB calibration can be established from allometry or from a small destructive sample, and the residual variance is an open question that TWIG's multi-temporal design is positioned to resolve.

References

Calders, K., Newnham, G., Burt, A., Murphy, S., Raumonen, P., Herold, M., Culvenor, D., Avitabile, V., Disney, M., Armston, J., Kaasalainen, M., 2015. Nondestructive estimates of above-ground biomass using terrestrial laser scanning. Methods in Ecology and Evolution 6, 198–208. https://doi.org/10.1111/2041-210X.12301

Chave, J., Réjou-Méchain, M., Búrquez, A., Chidumayo, E., Colgan, M.S., Delitti, W.B.C., Duque, A., Eid, T., Fearnside, P.M., Goodman, R.C., Henry, M., Martínez-Yrízar, A., Mugasha, W.A., Muller-Landau, H.C., Mencuccini, M., Nelson, B.W., Ngomanda, A., Nogueira, E.M., Ortiz-Malavassi, E., Pélissier, R., Ploton, P., Ryan, C.M., Saldarriaga, J.G., Vieilledent, G., 2014. Improved allometric models to estimate the aboveground

biomass of tropical trees. Global Change Biology 20, 3177–3190. https://doi.org/10.1111/gcb.12629

Stereńczak, K., Kulicki, M., Kraszewski, B., Gasica, T.A., Mitelsztedt, K., Mielcarek, M., Tangwa, E., Erfanifard, Y., Brach, M., Handayani, H.H., Ramadhani, A.N.R., Hawryło, P., Krok, G., Armazeta, A.J.G., Arisandy, D.R., Błaś, F., Damayanti, R.A., De Yong, R.J., Deden, Fabjan, K., Fathimah, F.N., Górska, J., Ila, Q.P.A.N., Irawan, R.D., Iwaniec, F., Jabłoński, A., Krawczyk, W., Kwaśniewski, B., Kwartnik, K., Mardhatillah, R., Massayudevi, R.A., Miszczyszyn, J., Murti, S.B.A., Nabila, D.A., Novatiana, R., Prasetyo, W.E., Pratama, F., Ramadhani, A.T.U., Rębisz, N., Rizannyansyah, H.F.F., Seremak, Z., Skomorucha, B., Skóra, D., Sukmawan, N., Taslyanto, C.A.P., Turemka, G., Valika, E., Warman, D.M., Wilczyńska, A., Winkowska, K., Witkowski, K., Yudisti, S.A., Zdunek, N., Zieliński, D., Ziółkowski, A., 2026. A Central European tree species dataset of annotated terrestrial laser scanning point clouds - TreeScanPL10K. Sci Data. https://doi.org/10.1038/s41597-026-07269-1

Taubin, G., 1991. Estimation of planar curves, surfaces, and nonplanar space curves defined by implicit equations with applications to edge and range image segmentation. IEEE Transactions on Pattern Analysis and Machine Intelligence 13, 1115–1138. https://doi.org/10.1109/34.103273